Topic P6: Radioactivity

P6.1 Radioactive emissions

Summary
Having considered the general characteristics of waves and particles, we now move on to look at radioactive decay which combines these two ideas. The idea of isotopes is introduced, leading into looking at the different types of emissions from atoms.

Underlying knowledge and understanding
Learners should have prior understanding of the atomic model, chemical symbols and formulae. An understanding of radioactivity is not expected and should be taught as new content.

Common misconceptions
Learners tend to struggle with the concept that radioactivity is a random and unpredictable process. The idea of half-life is another area that can lead to confusion. Learners often find it difficult to understand that objects being irradiated does not lead to them becoming radioactive.

Tiering
Statements shown in **bold** type will only be tested in the Higher Tier papers.
All other statements will be assessed in both Foundation and Higher Tier papers.

Topic content			Opportunities to cover:		
Learning outcomes		To include	Maths	Working scientifically	Practical suggestions
P6.1a	recall that atomic nuclei are composed of both protons and neutrons, that the nucleus of each element has a characteristic positive charge		M5b		
P6.1b	recall that atoms of the same elements can differ in nuclear mass by having different numbers of neutrons				
P6.1c	use the conventional representation for nuclei to relate the differences between isotopes	identities, charges and masses			

P6.1d	recall that some nuclei are unstable and may emit alpha particles, beta particles, or neutrons, and electromagnetic radiation as gamma rays			WS1.1a, WS1.1b, WS1.2a, WS1.2d, WS1.3b, WS1.3f	Use of a Geiger-Müller tube and radioactive sources to investigate activity.
P6.1e	relate the emission of alpha particles, beta particles, gamma radiation and neutrons to possible changes in the mass or the charge of the nucleus, or both				
P6.1f	use names and symbols of common nuclei and particles to write balanced equations that represent radioactive decay				
P6.1g	balance equations representing the emission of alpha, beta or gamma radiation in terms of the masses, and charges of the atoms involved		M1b, M1c, M3c		
P6.1h	recall that in each atom its electrons are arranged at different distances from the nucleus, that such arrangements may change with absorption or emission of electromagnetic radiation and that atoms can become ions by loss of outer electrons	knowledge that inner electrons can be 'excited' when they absorb energy from radiation and rise to a higher energy level. When this energy is lost by the electron it is emitted as radiation. When outer electrons are lost this is called ionisation			
P6.1i	recall that changes in atoms and nuclei can also generate and absorb radiations over a wide frequency range	an understanding that these types of radiation may be from any part of the electromagnetic spectrum which includes gamma rays		WS1.1b, WS1.3e	Demonstration of fluorescence with a black light lamp and tonic water.

P6.1j	explain the concept of half-life and how this is related to the random nature of radioactive decay		M1c, M3d, M4a, M4c	WS1.1b, WS1.3a, WS1.3b, WS1.3c, WS1.3e, WS1.3f, WS1.3h, WS2a	Using dice to model random decay and half-life. Research how half-life can be used in radioactive dating.
P6.1k	**calculate the net decline, expressed as a ratio, during radioactive emission after a given (integral) number of half-lives**	**half-life graphs**	M1c, M3d		
P6.1l	recall the differences in the penetration properties of alpha particles, beta particles and gamma rays			WS1.1b, WS1.2a, WS1.2b, WS1.2c, WS1.3a, WS1.3f, WS1.3g, WS1.3h	Use of Geiger-Müller tube, sources and aluminium plates of varying thicknesses to investigate change in count rate.